

Arjuna JEE 2025

Mathematics

DPP: 8

Permutations and Combinations

Q1 The number of ways in which a pack of 52 cards can be divided into 4 sets, three of them containing 17 cards and the fourth just one card is

- (A) $52!(17!)^3$
 (B) $\frac{52!}{3!(17!)^3}$
 (C) $\frac{52!}{4!(17!)^3}$
 (D) None of these

Q2 The number of ways in which a pack of 52 cards can be divided equally among four players on orders, is

- (A) $\frac{52!}{(13!)^4}$
 (B) $\frac{52!}{(13!)^4 4!}$
 (C) $\frac{52!}{13!}$
 (D) $\frac{52!}{4! \times 13!}$

Q3 The number of ways in which a pack of 52 cards can be divided equally into four groups, is

- (A) $\frac{52!}{(13!)^4}$
 (B) $\frac{52!}{(13!)^4 4!}$
 (C) $\frac{52!}{(13!)^4 \times 4}$
 (D) $\frac{52!}{13! \times 4!}$

Q4 In how many ways can 10 balls be divided between two boys, one receiving two and other eight balls, is

- (A) 45
 (B) 75
 (C) 90
 (D) none of these

Q5

The number of ways in which 9 students can be equally distributed among 3 sections, is

- (A) $\frac{9!}{3(3!)^2}$
 (B) $\frac{9!}{(3!)^3}$
 (C) $9!$
 (D) none of these

Q6 In how many ways can 8 different books be distributed among 3 students if each receives at least 2 books?

- (A) 2980
 (B) 3540
 (C) 2940
 (D) 1260

Q7 The total number of ways of dividing 15 different things into groups of 8, 4 and 3 respectively, is

- (A) $\frac{(15!)}{8!.4!.3!.3!}$
 (B) $\frac{(15!)}{8!.4!.3!}$
 (C) $\frac{(15)!}{8!.4!.2!}$
 (D) None of these

Q8 The number of ways in which 6 different toys may be distributed among 4 children so that each child gets at least one toy is

- (A) ${}^6P_4 \times 2^4$
 (B) ${}^6C_4 \times 2^4$
 (C) 1560
 (D) None of these

Q9 The number of ways 10 boys can be divided into 2 groups of 5, such that the two tallest boys are in the different groups, is

- (A) 70
 (B) 35



(C) 252

(D) 126

Q10 The number of ways in which 200 different things can be divided into groups of 100 pairs, is

(A) $\frac{(200)!}{2^{100}}$

(B) $\frac{(200)!}{(100)! \cdot 2^{100}}$

(C) $(101) \cdot (102) \dots (200)$

(D) none of these

Q11 In how many ways can 9 different books be distributed among three students if each receives atleast 2 books.

(A) 11465

(B) 10506

(C) 11508

(D) 10508

Q12 Find the prime factorization of $10!$, i.e. if $10! = 2^p \cdot 3^q \cdot 5^r \cdot 7^s$, then find p, q, r and s .

Q13 In how many ways can 17 persons depart from railway station in 2 cars and 3 autos, given that 2 particular persons depart by the same car (4 persons can sit in a car and 3 persons can sit in an auto)?

(A) $\frac{15!}{2!4!(3!)^3}$

(B) $\frac{16!}{(2!)^2 4!(3!)^3}$

(C) $\frac{17!}{2!4!(3!)^3}$

(D) $\frac{15!}{4!(3!)^3}$

Q14 Given five different green dyes, four different blue dyes and three different red dyes, how many combination of dyes can be chosen taking at least one green, one blue dye?

(A) 2880

(B) 3200

(C) 3630

(D) 3720

Q15 The number of divisors of $a^p b^q c^r d^s$, where a, b, c, d are primes and $p, q, r, s \in \mathbb{N}$ excluding 1 and the number itself, is

(A) $pqr s$

(B) $(p+1)(q+1)(r+1)(s+1) - 4$

(C) $pqr s - 2$

(D) $(p+1)(q+1)(r+1)(s+1) - 2$

Q16 Number of ways in which 9 different toys be distributed among 4 children belonging to different age group in such a way that distribution among the 3 elder children is even and the youngest one is to receive one toy more, is

(A) $\frac{(5!)^2}{8}$

(B) $\frac{9!}{2}$

(C) $\frac{9!}{3!(2!)^3}$

(D) None of these

Q17 There are 3 different books of mathematics, 4 different of science and 5 different of English. How many different collections can be made such that each collection consists of one book of each subject

(A) 60

(B) 30

(C) 47

(D) 80

Q18 There are 3 different books of mathematics, 4 different of science and 5 different of English. How many different collections can be made such that each collection consists of: at least one book of each subject

(A) 3155

(B) 3255

(C) 3048

(D) 2180

Q19 There are 3 different books of mathematics, 4 different of science and 5 different of English. How many different collections can be made such that each collection consists of at least one book of English

(A) 3948

(B) 3987

(C) 3947

(D) 3968


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Q20 There are 4 oranges, 5 apples and 6 mangoes in a fruit basket. In how many ways can a person make a selection of fruits from among the fruits in the basket, if All fruits of the same type are identical

- (A) 209 (B) $2^{15} - 1$
(C) 2^{10} (D) 210

Q21 Direction: There are 4 oranges, 5 apples and 6 mangoes in a fruit basket. In how many ways can a person make a selection of fruits from among the fruits in the basket, if All fruits of the same type are different

- (A) 209 (B) $2^{15} - 1$
(C) 2^{10} (D) 210



Answer Key

Q1 (B)

Q2 (A)

Q3 (B)

Q4 (C)

Q5 (B)

Q6 (C)

Q7 (B)

Q8 (A)

Q9 (A)

Q10 (B)

Q11 (C)

Q12 $p=8, q=4, r=2, s=1$

Q13 (D)

Q14 (D)

Q15 (D)

Q16 (C)

Q17 (A)

Q18 (B)

Q19 (D)

Q20 (A)

Q21 (B)

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